

Mechanics

Lecture 9
Rigid Body Motion
(Chapter 4, 5)

What We Did Last Time

- Discussed 3-dimensional rotation
 - Preparation for rigid body motion
 - Movement in 3-d + Rotation in 3-d = 6 coordinates
- Looked for ways to describe 3-d rotation
 - Euler angles one of the many possibilities
 - Euler's theorem
- Defined infinitesimal rotation $d\Omega$
 - Commutative (unlike finite rotation)
 - Behaves as an axial vector (like angular momentum)

$$d\Omega = \mathbf{n} d\Phi$$

$$d\mathbf{r} = \mathbf{r} \times d\Omega$$

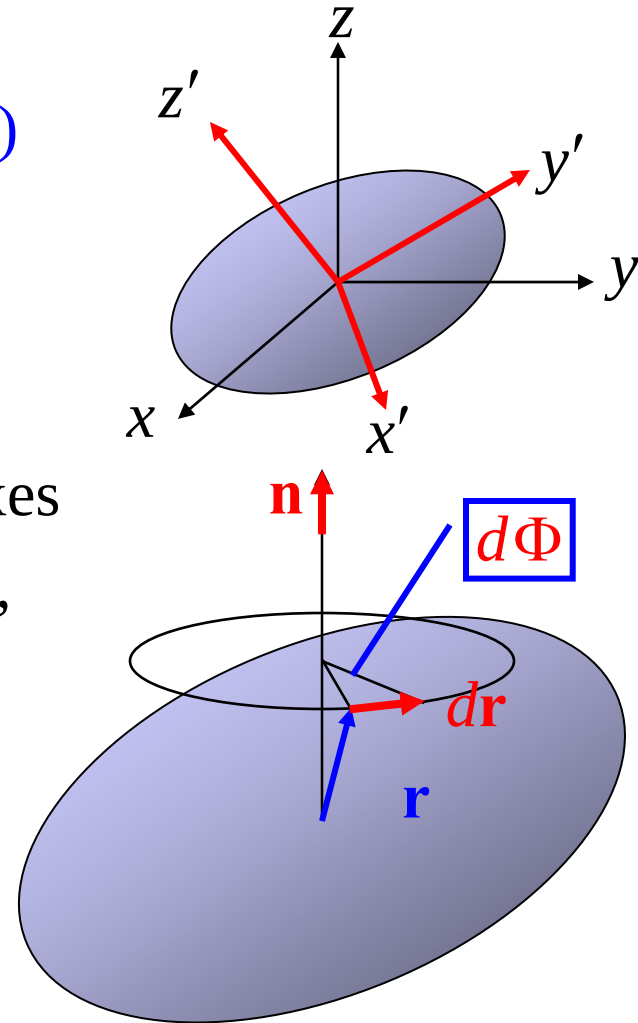
Goals For Today

- Time derivatives in rotating coordinate system
 - To calculate velocities and accelerations
 - Coriolis effect
 - Express angular velocity using Euler angles
- Try to write down Lagrangian for rigid body
 - Separate rotation from movement of CoM
 - Define inertia tensor
- Will *almost* get to the equation of motion

Body Coordinates

- Consider a rotating rigid body
 - Define **body coordinates** (x', y', z')
- Between t and $t + dt$, the body coordinates rotate by $d\Omega = \mathbf{n}d\Phi$
 - The direction is **CCW** because we are talking about the coordinate axes
 - Observed in the space coordinates, any point $\mathbf{r}$ on the body moves by $d\mathbf{r} = d\Phi \mathbf{n} \times \mathbf{r} = d\Omega \times \mathbf{r}$

Sign opposite from last lecture
because the rotation is CCW



General Vectors

- Now consider a general vector $\mathbf{G}$
 - How does it move in space/body coordinates?
 - i.e. what's the time derivative $d\mathbf{G}/dt$?
- Movement $d\mathbf{G}$ differs in space and body coordinates because of the rotation of the latter

$$(d\mathbf{G})_{\text{space}} = (d\mathbf{G})_{\text{body}} + (d\mathbf{G})_{\text{rot}}$$

Difference is due to rotation

- If $\mathbf{G}$ is fixed to the body

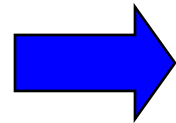
$$(d\mathbf{G})_{\text{body}} = 0 \quad \text{and} \quad (d\mathbf{G})_{\text{space}} = d\boldsymbol{\Omega} \times \mathbf{G}$$

$$\Rightarrow (d\mathbf{G})_{\text{rot}} = d\boldsymbol{\Omega} \times \mathbf{G}$$

Generally true

Angular Velocity

- For any vector $\mathbf{G}$ $(d\mathbf{G})_{\text{space}} = (d\mathbf{G})_{\text{body}} + d\mathbf{\Omega} \times \mathbf{G}$



$$\left(\frac{d\mathbf{G}}{dt}\right)_{\text{space}} = \left(\frac{d\mathbf{G}}{dt}\right)_{\text{body}} + \boldsymbol{\omega} \times \mathbf{G}$$

$$\boldsymbol{\omega} dt = d\mathbf{\Omega} = d\Phi \mathbf{n}$$

- $\boldsymbol{\omega}$ = instantaneous angular velocity
 - Direction = $\mathbf{n}$ = instantaneous axis of rotation
 - Magnitude = $d\Phi/dt$ = instantaneous rate of rotation
- Since this works for any vector, we can say

$$\left(\frac{d}{dt}\right)_s = \left(\frac{d}{dt}\right)_r + \boldsymbol{\omega} \times$$

space coordinates
rotating coordinates

Coriolis Effect

$$\left(\frac{d}{dt}\right)_s = \left(\frac{d}{dt}\right)_r + \boldsymbol{\omega} \times$$

- Imagine a particle observed in a rotating system

- e.g. watching an object's motion on Earth

- Velocity: $\mathbf{v}_s = \mathbf{v}_r + \boldsymbol{\omega} \times \mathbf{r}$

- Acceleration:
$$\mathbf{a}_s = \left(\frac{d\mathbf{v}_s}{dt}\right)_s = \left(\frac{d\mathbf{v}_s}{dt}\right)_r + \boldsymbol{\omega} \times \mathbf{v}_s$$
$$= \mathbf{a}_r + 2\boldsymbol{\omega} \times \mathbf{v}_r + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

- Newton's equation works in the space (inertial) system,
i.e. $\mathbf{F} = m\mathbf{a}_s$

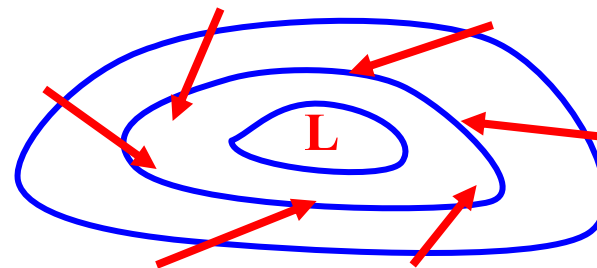
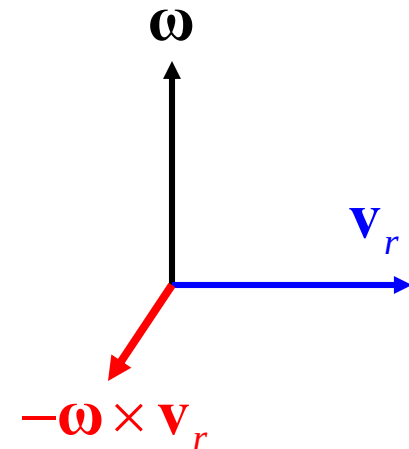
➡ $m\mathbf{a}_r = \mathbf{F}_{\text{eff}} = \mathbf{F} - 2m(\boldsymbol{\omega} \times \mathbf{v}_r) - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$

Object appears to move according to this force

Coriolis Effect

$$m\mathbf{a}_r = \mathbf{F}_{\text{eff}} = \mathbf{F} - 2m(\boldsymbol{\omega} \times \mathbf{v}_r) - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

- Last term is centrifugal force
- Middle term is Coriolis effect
 - Occurs when object is moving in the rotating frame
- Moving objects on Earth appears to deflect toward right in the northern hemisphere
 - Hurricane wind pattern
 - Foucault pendulum



Coriolis Effect

- Free-falling particle
 - Drop a ball and ignore air resistance
 - Coriolis effect is $-2m(\boldsymbol{\omega} \times \mathbf{v})$
 - Pointing east (out of screen) $\rightarrow$ x axis
 - Ignoring the centrifugal force

$$\begin{cases} m\ddot{x} = -2m\omega v_z \sin \theta \\ m\ddot{z} = -mg \end{cases}$$

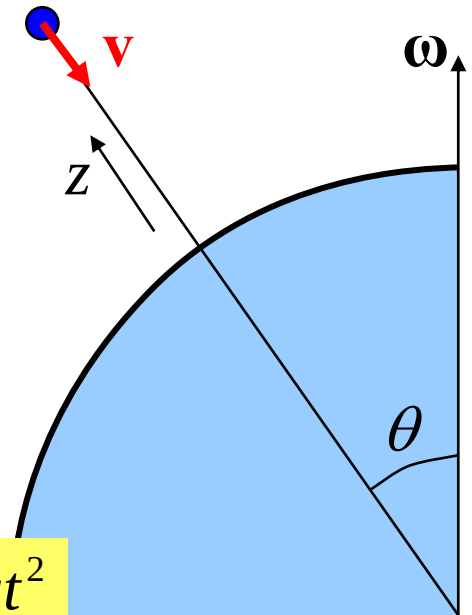


$$x = \frac{\omega g t^3 \sin \theta}{3}, \quad z = -\frac{g t^2}{2}$$



$$x = \frac{\omega \sin \theta}{3} \sqrt{\frac{(-2z)^3}{g}}$$

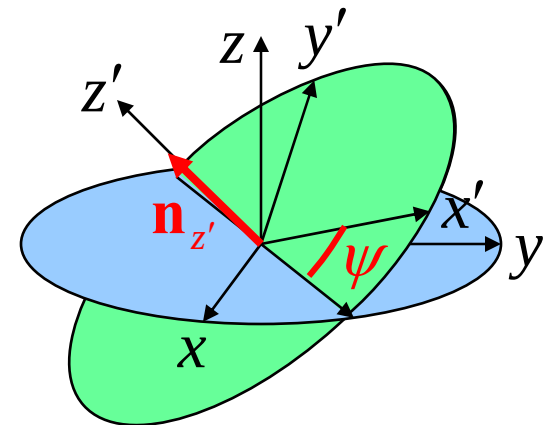
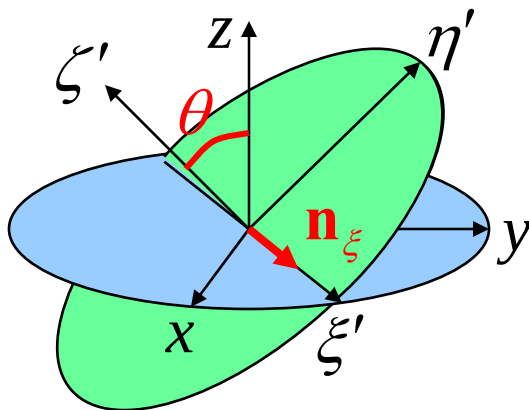
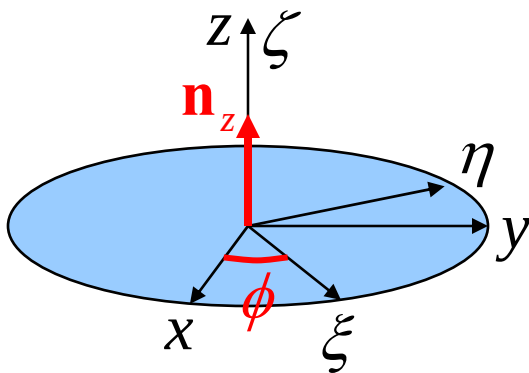
- For $\sin \theta = 1$ and $z = 100 \text{ m} \rightarrow x = 2.2 \text{ cm}$



Euler Angles

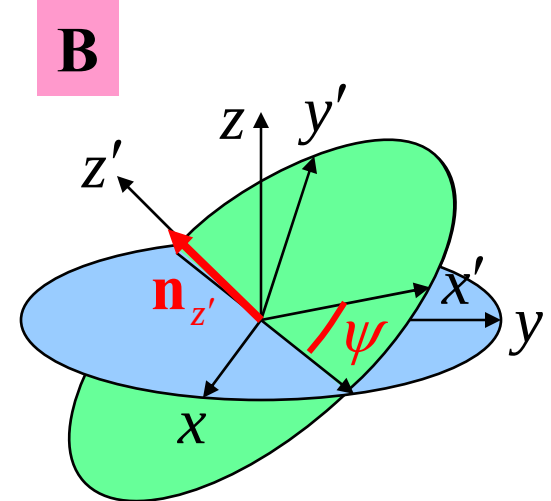
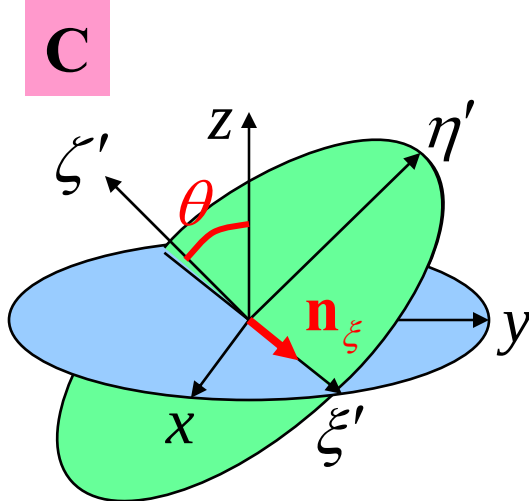
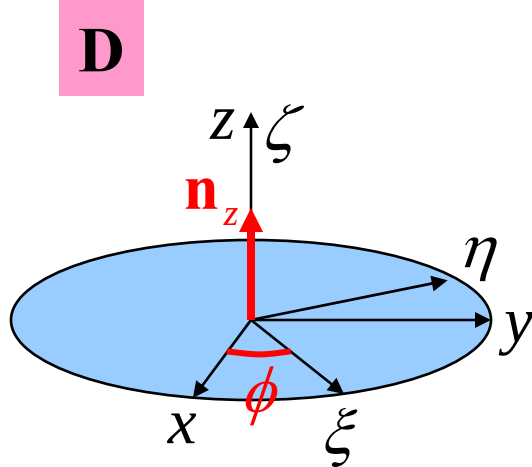
- Use angular velocity ω to calculate particles' velocities
- Use Euler angles to describe the rotation of rigid bodies
- How are they connected?
 - Infinitesimal rotations can be added like vectors

$$\boldsymbol{\omega} = \mathbf{n}_z \dot{\phi} + \mathbf{n}_\xi \dot{\theta} + \mathbf{n}_{z'} \dot{\psi}$$



Euler Angles

- Let's express ω in $x'-y'-z'$
 - Doing this in $x-y-z$ equally easy (or difficult)
 - Must express $\mathbf{n}_z, \mathbf{n}_\xi, \mathbf{n}_\eta$ in $x'-y'-z'$
 - $\mathbf{n}_z \rightarrow \mathbf{A}\mathbf{n}_z, \mathbf{n}_\xi \rightarrow \mathbf{B}\mathbf{n}_\xi$ ($\mathbf{n}_\eta$ is OK)



Euler Angles

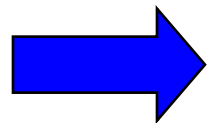
■ From the last lecture

$$\mathbf{D} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} \cos \psi \cos \phi - \cos \theta \sin \phi \sin \psi & \cos \psi \sin \phi + \cos \theta \cos \phi \sin \psi & \sin \psi \sin \theta \\ -\sin \psi \cos \phi - \cos \theta \sin \phi \cos \psi & -\sin \psi \sin \phi + \cos \theta \cos \phi \cos \psi & \cos \psi \sin \theta \\ \sin \theta \sin \phi & -\sin \theta \cos \phi & \cos \theta \end{bmatrix}$$



$$\mathbf{n}_z = \mathbf{A} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sin \psi \sin \theta \\ \cos \psi \sin \theta \\ \cos \theta \end{bmatrix}$$

$$\mathbf{n}_\xi = \mathbf{B} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos \psi \\ -\sin \psi \\ 0 \end{bmatrix}$$

$$\mathbf{n}_{z'} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Euler Angles

$$\boldsymbol{\omega} = \mathbf{n}_z \dot{\phi} + \mathbf{n}_\xi \dot{\theta} + \mathbf{n}_{z'} \dot{\psi} = \begin{bmatrix} \dot{\phi} \sin \psi \sin \theta + \dot{\theta} \cos \psi \\ \dot{\phi} \cos \psi \sin \theta - \dot{\theta} \sin \psi \\ \dot{\phi} \cos \theta + \dot{\psi} \end{bmatrix}$$

- Remember: this is in the x' - y' - z' system
- Now we know how to express velocities in terms of time-derivatives of Euler angles
 - We can write down the Lagrangian

Kinetic Energy

- Kinetic energy of multi-particle system is

$$T = \frac{1}{2} M v^2 + \frac{1}{2} m_i v_i'^2$$

Remember Einstein convention

Motion of CoM

Motion around CoM

- If we define the body axis from the center of mass

$$T = \frac{1}{2} M (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + T'(\dot{\phi}, \dot{\theta}, \dot{\psi})$$

- T' depends only on the angular velocity
- Must be a 2nd order homogeneous function

Potential Energy

- Potential energy can often be separated as well

$$V = V_1(x, y, z) + V_2(\phi, \theta, \psi)$$

- Uniform gravity $\mathbf{g}$ $V_1 = -\mathbf{g} \cdot \mathbf{r}$

- Uniform magnetic field $\mathbf{B}$ and magnetic dipole moment $\mathbf{M}$

$$V_2 = -\mathbf{M} \cdot \mathbf{B}$$

- Lagrangian can then be written as

$$L = L_t(x, y, z, \dot{x}, \dot{y}, \dot{z}) + L_r(\phi, \theta, \psi, \dot{\phi}, \dot{\theta}, \dot{\psi})$$



Translational



Rotational

It is often possible to separate the translational and rotational motions by taking the center of mass as the origin of the body coordinate axes

Rotational Motion

- We concentrate on the rotational part
 - Translational part same as a single particle → Easy
- Consider total angular momentum $\mathbf{L} = m_i \mathbf{r}_i \times \mathbf{v}_i$
 - $\mathbf{v}_i$ is given by the rotation $\boldsymbol{\omega}$ as $\mathbf{v}_i = \boldsymbol{\omega} \times \mathbf{r}_i$

$$\mathbf{L} = m_i \mathbf{r}_i \times (\boldsymbol{\omega} \times \mathbf{r}_i)$$

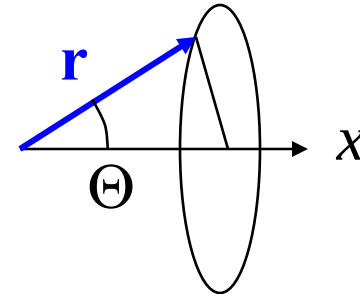
$$= m_i \left[\boldsymbol{\omega} r_i^2 - \mathbf{r}_i (\mathbf{r}_i \cdot \boldsymbol{\omega}) \right] = \begin{bmatrix} m_i(r_i^2 - x_i^2) & -m_i x_i y_i & -m_i x_i z_i \\ -m_i y_i x_i & m_i(r_i^2 - y_i^2) & -m_i y_i z_i \\ -m_i z_i x_i & -m_i z_i y_i & m_i(r_i^2 - z_i^2) \end{bmatrix} \boldsymbol{\omega}$$

Inertia tensor $\mathbf{I}$

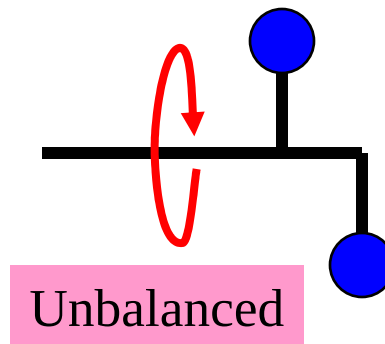
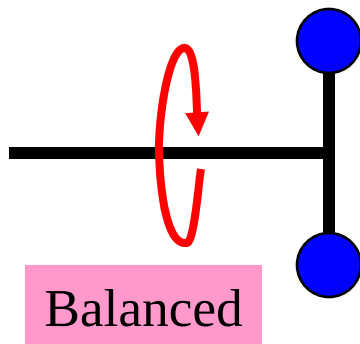
Inertia Tensor

- Diagonal components are familiar moment of inertia

$$I_{xx} = m_i(r_i^2 - x_i^2) = m_i r_i^2 \sin^2 \Theta$$



- What are the off-diagonal components?
 - I_{yx} produces L_y when the object is turned around x axis
 - Imagine turning something like:



Unbalanced one has non-zero off-diagonal components, which represents “wobbliness” of rotation

Inertia Tensor

- Using $(x_i, y_i, z_i) \rightarrow (x_{i1}, x_{i2}, x_{i3})$

$$\mathbf{I} = \begin{bmatrix} m_i(r_i^2 - x_i^2) & -m_i x_i y_i & -m_i x_i z_i \\ -m_i y_i x_i & m_i(r_i^2 - y_i^2) & -m_i y_i z_i \\ -m_i z_i x_i & -m_i z_i y_i & m_i(r_i^2 - z_i^2) \end{bmatrix} \Rightarrow I_{jk} = m_i (r_i^2 \delta_{jk} - x_{ij} x_{ik})$$

- We can also deal with continuous mass distribution $\rho(\mathbf{r})$

$$I_{jk} = \int \rho(\mathbf{r}) (r^2 \delta_{jk} - x_j x_k) d\mathbf{r}$$

Kinetic Energy

$$I_{jk} = m_i (r_i^2 \delta_{jk} - x_{ij} x_{ik})$$

- Kinetic energy due to rotation is $T = \frac{1}{2} m_i \mathbf{v}_i \cdot \mathbf{v}_i$
 - Using $\mathbf{v}_i = \boldsymbol{\omega} \times \mathbf{r}_i$

$$T = \frac{1}{2} m_i \mathbf{v}_i \cdot (\boldsymbol{\omega} \times \mathbf{r}_i) = \frac{\boldsymbol{\omega}}{2} \cdot m_i (\mathbf{r}_i \times \mathbf{v}_i) = \frac{\boldsymbol{\omega}}{2} \cdot \mathbf{L}$$

- Using $\mathbf{L} = \mathbf{I} \boldsymbol{\omega}$ $\Rightarrow T = \frac{\boldsymbol{\omega} \cdot \mathbf{I} \boldsymbol{\omega}}{2}$

- Defining $\mathbf{n}$ as the unit vector in the direction of $\boldsymbol{\omega}$ $\boldsymbol{\omega} = \omega \mathbf{n}$

$$T = \frac{\mathbf{n} \cdot \mathbf{I} \mathbf{n}}{2} \omega^2 = \frac{1}{2} I \omega^2 \quad \text{where} \quad I = \mathbf{n} \cdot \mathbf{I} \mathbf{n} = m_i [r_i^2 - (\mathbf{r}_i \cdot \mathbf{n})^2]$$

Moment of inertia about axis $\mathbf{n}$

- **NB:** $\mathbf{n}$ moves with time
- $I = I(t)$ must change accordingly with time

Shifting Origin

- Origin of body axes does not have to be at the CoM
 - It's convenient – Separates translational/rotational motion
- If it isn't, I can be easily translated

$$\begin{aligned} & \boxed{\text{from origin}} \rightarrow \mathbf{r}_i = \mathbf{R} + \mathbf{r}'_i \leftarrow \boxed{\text{from CoM}} \\ & I = m_i (\mathbf{r}_i \times \mathbf{n})^2 = m_i [(\mathbf{R} + \mathbf{r}'_i) \times \mathbf{n}]^2 \\ & = M (\mathbf{R} \times \mathbf{n})^2 + m_i (\mathbf{r}'_i \times \mathbf{n})^2 + \cancel{2m_i (\mathbf{R} \times \mathbf{n}) \cdot (\mathbf{r}'_i \times \mathbf{n})} \end{aligned}$$

$\swarrow$ $\boxed{I \text{ of CoM}}$ $\nwarrow$ $\boxed{I \text{ from CoM}}$

Summary

- Found the velocity due to rotation

- Used it to find Coriolis effect

$$\left(\frac{d}{dt}\right)_s = \left(\frac{d}{dt}\right)_r + \boldsymbol{\omega} \times$$

- Connected $\boldsymbol{\omega}$ with the Euler angles

- Lagrangian $\rightarrow$ translational and rotational parts

- Often possible if body axes are defined from the CoM

- Defined the inertia tensor

$$\mathbf{I} = m_i \left(r_i^2 \delta_{jk} - x_{ij} x_{ik} \right)$$

- Calculated angular momentum and kinetic energy

$$\mathbf{L} = \mathbf{I}\boldsymbol{\omega}$$

$$T = \frac{\mathbf{nI}\mathbf{n}}{2} \omega^2 = \frac{1}{2} I \omega^2$$

- Next: Equation of motion (finally!)